Is the closing-auction imbalance already in the price? Evidence from 5.2 million Nasdaq auction snapshots

Group name	Closing Cross
Dataset	kaggle.com/competitions/optiver-trading-at-the-close (file `train.csv`)
Code	github.com/USERNAME/nasdaq-closing-cross (code, README and supplement)

#	Name	Roll no.	Institute email	Contribution
1	Arush Gumber	2023136	arush23136@iiitd.ac.in	Problem statement, Tests 3 to 5, code (4, 5)
2	Aditri Jain	2024027	aditri24027@iiitd.ac.in	Dataset and size check (1)
3	Arnesh Batra	2023129	arnesh23129@iiitd.ac.in	Preparation, before and after (2, 3)
4	Anika Goyal	2024072	anika24072@iiitd.ac.in	EDA, Tests 1 and 2 (4, 5)
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1. Dataset

Source: Optiver, "Trading at the Close", a Kaggle competition released in 2023. It records the last ten minutes of the Nasdaq closing auction, the call auction that sets each stock's official closing price at 16:00, for 200 stocks over 481 trading days. One row is one stock at one 10-second snapshot between 15:50 and 15:59.

What is raw about it: the indicative clearing prices are missing for half the rows by exchange rule; four stock-days carry no book at all; far prices reach hundreds of times the reference price; 15 stocks leave and re-enter the panel; the response is a 60-second move sampled every 10 seconds, so consecutive rows share most of their window; and the index that the response is measured against is hidden.

	Raw file	After encoding
Instances (rows)	5,237,980	5,237,760
Attributes (columns)	17	223
Instances $\times$ attributes	89,045,660	1,168,020,480

Size check: the raw file alone gives $5{,}237{,}980 \times 17 = 89{,}045{,}660 \geq 3{,}000{,}000$. After cleaning we keep 13 numeric columns, add 5 engineered ones and one-hot encode the stock (200 levels), the imbalance side (3) and the auction phase (2), which gives $5{,}237{,}760 \times 223 = 1{,}168{,}020{,}480$.

Key features

Feature	Type	Real-world meaning
`target`	continuous	the stock's 60 s return minus the index's 60 s return, in basis points (1 bp $= 0.01\%$); our response
`imbalance_size`	continuous	shares that would be left unmatched if the auction cleared now
`imbalance_buy_sell_flag`	nominal	side of that imbalance: buy, sell or none
`matched_size`	continuous	shares that would trade at the current reference price
`reference_price`	continuous	the price at which the auction would clear now
`near/far_price`	continuous	indicative clearing prices with and without the continuous book; published from 15:55
`bid/ask_price`	continuous	best quotes in the continuous market
`seconds_in_bucket`	discrete	seconds since 15:50 (0 to 540)
`signed_imbalance`	engineered	side $\times$ imbalance / matched shares: $+0.3$ means buyers want 30% more shares than are matched, -0.3 the same for sellers

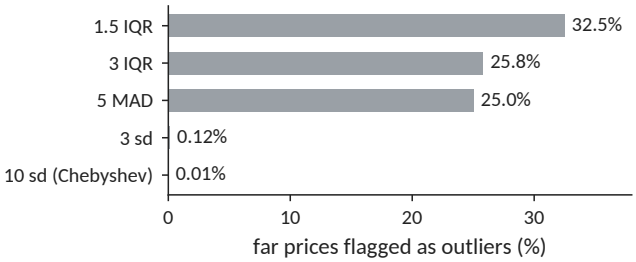


Figure 1: Quartile-based fences would discard a quarter of all far prices. The Chebyshev fence removes only physically impossible values. Bars show the share of far prices each rule calls an outlier.

2. Data preparation

Every step below was fixed before any hypothesis test was run, and each records its row counts.

Issue found	What we did	Why, and not the alternative
220 rows (4 stock-days) with no book at all	dropped	no market existed for that stock that day; 88 of these rows also lack the target, so imputing would invent both
Near and far price missing in 2,856,960 rows before 15:55	kept missing; added a phase flag	the exchange publishes them only from 15:55. Imputing would hand the analysis information no trader had
Far prices hundreds of times the reference	207 values beyond 10 sd set to missing	Tukey and MAD fences flag a quarter of all far prices (Figure 1), because the far price spreads out as the imbalance grows: that is economics, not error. Chebyshev guarantees a 10 sd fence flags at most 1% under any distribution
Far price missing in 37,162 rows after 15:55	kept missing	it needs auction-only interest on both sides; no test uses it
Target with excess kurtosis 22.6	winsorised copy at 0.1% and 99.9% (-49.4 and 51.8 bps); raw kept	extreme moves are real in an auction, so no row is deleted. Mean-based tests use the clipped copy and report the raw result beside it
60 s targets sampled every 10 s	tests use one snapshot per minute	neighbouring targets share up to 50 of their 60 seconds (Figure 2); every row would inflate the sample about sixfold
Hidden index behind the target	recovered by least squares	the implied index return is the same for every stock at a timestamp (median spread 0.005 bps), and 200 positive weights summing to 1.0000 reproduce it with unexplained variance 2.1×10^{-8}. The response is what it claims to be
Sizes and prices not comparable across stocks	engineered signed imbalance, spread and near/far gaps in bps	a 1M-share imbalance is large for a small stock and noise for a large one
Side coded $-1, 0, 1$; 200 stock ids	one-hot encoded	an integer code would make "sell $<$ none $<$ buy" a magnitude
Duplicate keys	checked on (stock, day, second)	none found

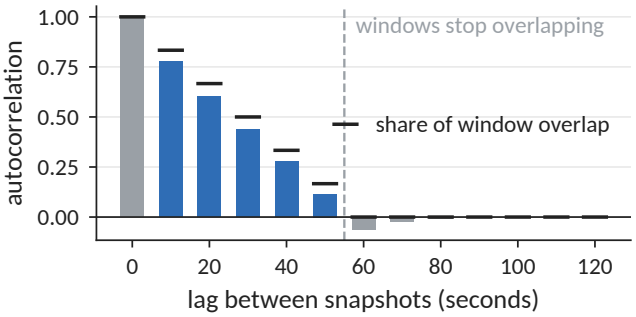


Figure 2: Consecutive targets are correlated only because their windows overlap. Bars are the correlation between a target and the one k seconds later; black ticks are the share of the 60 s window the two have in common. Both reach zero at 60 s, so tests use one snapshot per minute.

Processing at stream scale

To total the imbalance of each of the 95,232 stock-days in one pass, as a live feed would require, we streamed every row through a Count-Min sketch with 15% of the memory of exact counting. Every estimate stayed within the sketch's εN guarantee, and 9 of the 10 heaviest stock-days were found.

3. What changed after preparation

Measure	Before	After
Rows	5,237,980	5,237,760
Stock-days present (of 96,200)	95,236	95,232
Missing cells	5,752,930	5,751,289 (5,713,920 by exchange rule)
Impossible far prices	207	0
Memory	934 MB	400 MB
Rows used per test	every snapshot	952,320 (one per minute)

Cleaning removed almost no data: only 220 rows went. What changed is the unit of analysis. The autocorrelation of the target follows the overlap of its windows and vanishes at exactly one minute (Figure 2), so the honest sample is one snapshot per minute. Across days, 968 stock-days are absent; one stock is missing for 295 days, so every comparison is made within a day.

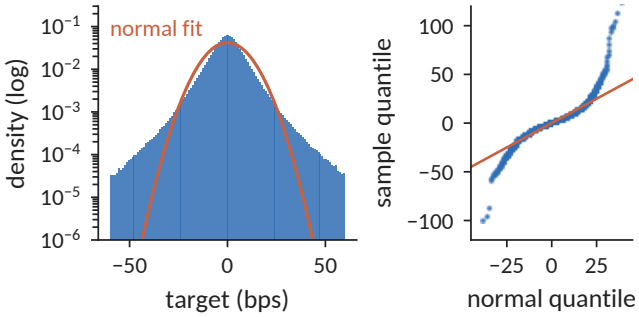


Figure 3: The target is far more peaked and heavy-tailed than a normal. Left: histogram on a log scale, with the normal of the same mean and spread in orange; the bars sit far above it in both tails. Right: Q-Q plot; normal data would lie on the line. So tests average to one value per day and rely on the CLT across days.

4. Basic questions and problem statements

Basic questions

1. **How lopsided are auctions?** Sell imbalances appear in 40% of snapshots, buy in 39%, and 22% are balanced.
2. **How costly is trading near the close?** The median quoted spread is 3.69 bps, and it narrows as 16:00 approaches (Figure 4).
3. **Is the target normal?** No (Figure 3). Moves beyond 3 sd occur in 1.49% of rows, 5.5 times the normal 0.27% but inside Chebyshev's limit of 11.1%. A Student t fitted by maximum likelihood has only 3.1 degrees of freedom: the variance exists, so the CLT and Chebyshev apply, but the fourth moment does not. It cuts the χ^2 goodness-of-fit statistic from 98,064 (normal) to 568, yet is still rejected at this sample size.
4. **Do imbalances change side at random?** No. A stock flips side 2.69 times a day on average, but the variance is 2.45 times the mean, and 19.6% of stock-days never flip against 6.8% under a Poisson; $\chi^2 = 189{,}527$ on 10 d.f. Some auctions are one-sided all along, others churn.

Figure 4 points to 15:55. At that moment Nasdaq starts publishing the near and far indicative prices, and it stops accepting new market-on-close orders. Balanced auctions rise from 13% to 25% of stocks within ten seconds.

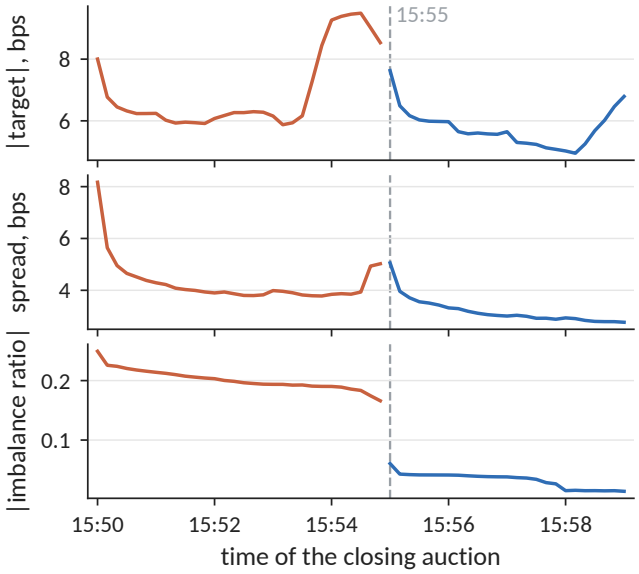


Figure 4: The auction changes regime at 15:55. Each line is an average over all stocks and days at each 10 s snapshot. Top: size of the next-minute move. Middle: bid-ask spread. Bottom: imbalance as a share of matched shares. The imbalance collapses at 15:55; the largest moves come from windows that contain 15:55.

Our problem statements

P1 (primary), information. Does the published imbalance predict the stock's next-minute move against the index?

P2, learned signal. Does a neural network trained on the whole auction state predict that move better than the imbalance alone?

P3, disclosure rule. Does the 15:55 rule change make the next minute more predictable from the imbalance, and less volatile?

P4, tradeability. Is any of this edge larger than the half-spread a trader pays to act on it?

PS	Tests	What links it to the others
P1	Tests 1, 2	establishes that the signal exists
P2	Test 3	asks whether learning finds more of the same signal
P3	Tests 4, 5	asks *when* the signal is strongest
P4	Test 6	asks whether any of P1 to P3 survives the cost of trading

How they are novel: the basic questions read one column at a time. Ours ask whether public information is already in the price, which needs an engineered signal, a day-level unit that respects shared index moves, and an economic yardstick (the spread). P2 tests the claim, common in recent work on learned trading signals, that deep models beat simple signals, with held-out days and confidence intervals. P3 compares the same stocks on the same day around a rule change. The public work on this dataset that we found fits predictive models; it does not test pricing, account for the overlapping targets or use the 15:55 boundary.

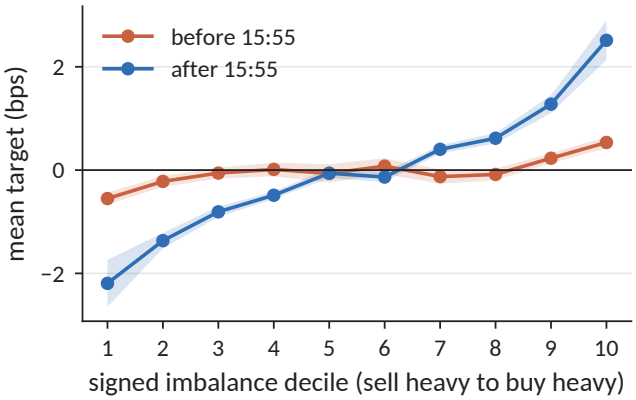


Figure 5: After 15:55 the imbalance sorts the next minute; before, it barely does. Every minute snapshot is ranked by signed imbalance and cut into ten equal groups, from the most sell-heavy (1) to the most buy-heavy (10). Each point is the group's mean move against the index. A fully priced imbalance would give a flat line at zero. Bands are 95% intervals over days.

5. Statistical inferences

Rows tested: the 952,320 snapshots at whole minutes (0, 60, …, 540 s), so no two targets of a stock overlap. Pre-15:55 rows are those at 0 to 240 s, whose windows end by 15:55; post rows are those at 300 to 540 s. The unit of every primary test is the trading day ($n = 481$), because all stocks on a day share one index move.

Smallest difference that matters (SDM): the median quoted half-spread, 1.85 bps. The cheapest way to act on the imbalance is to cross half the spread in the continuous book and exit in the closing cross, so a smaller edge cannot be traded. Each test states its own threshold; all were fixed before testing.

Reading a result: an effect *matters* when its whole 95% interval lies beyond the threshold, and *does not matter* when the whole interval lies inside it. We use $\alpha = 0.05$ per test and report Holm-adjusted p-values across the six.

Test 1 (P1): does the imbalance side predict the move?

a) Hypotheses, written before running the test

H_0	$\mu \le 0$
H_1	$\mu > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	one-sample t-test on day means
Smallest difference that matters	1.85 bps (the SDM)

μ is the expected edge per trade, $E[\text{side} \times \text{target}]$, for a trader who buys on a buy imbalance and sells on a sell imbalance. Each day gives one mean over its flagged snapshots. One-sided, because an efficient market predicts zero and liquidity demand predicts a positive edge.

b) Other tests we considered

Day means are independent across days and, by the CLT, close to normal at $n = 481$ even though rows are heavy-tailed; σ is unknown, so a one-sample t-test fits.

Also considered	Why we did not use it
z-test on all rows	treats 738,083 rows as independent although each day shares one index move; it gives $z = 42.0$, overstating the evidence. The 4,106,317 overlapping rows would be worse still
Welch t, buy vs sell rows	same question ($t = 41.9$, gap 0.93 bps) with the same row dependence; an F-test (ratio 1.09) rules out the pooled t
Bootstrap over days	no normality needed; outside the syllabus, so a check: 0.44 to 0.49 bps, same as the t interval
Permutation within snapshots	shuffles sides across stocks at each snapshot, 2,000 times; no shuffle came near the observed edge (largest 0.039 bps), p = 5.0×10^{-4}

c) Result

Estimate	$+0.47$ bps per trade (raw target: $+0.48$)
95% confidence interval	0.44 to 0.49 bps
Test statistic (d.f.)	$t = 32.5$ (480)
p-value (Holm)	8.0×10^{-124} (4.0×10^{-123})
Decision	reject H_0

The imbalance side predicts the next minute, on 458 of 481 days. The edge does not matter: even the top of its interval is 27% of the half-spread, so the information is in the price to within trading costs.

Test 2 (P1): is the direction of the move independent of the side?

a) Hypotheses, written before running the test

H_0	side and direction are independent
H_1	they are dependent
Significance level	$\alpha = 0.05$
Side	not applicable (χ^2 is one-tailed by construction)
Test	χ^2 test of independence, 3×2 table
Smallest difference that matters	Cramér's $V = 0.10$

Test 1 asks about the size of the move; Test 2 asks only about its sign, which is robust to the heavy tails. We use the first snapshot of each stock-day (95,193 rows), so each stock appears once per day. $V = 0.10$ is the conventional floor for a small association.

b) Other tests we considered

Also considered	Why we did not use it
Two-proportion z-test	drops the balanced rows; on buy vs sell it gives $z = 5.73$ and an up-share gap of 2.0 points (95% CI 1.3 to 2.7)
G-test	same large-sample limit; $G = 33.2$, same decision
χ^2 on all 951,770 minute rows	repeats each stock nine more times a day, so rows are not independent; $\chi^2 = 1{,}233$
Paired t on daily gaps	dependence-robust check: $t = 3.62$, p = 1.6×10^{-4}

c) Result

Estimate	$V = 0.019$; up-share 50.4% after buy vs 48.4% after sell
Test statistic (d.f.)	$\chi^2 = 33.2$ (2)
p-value (Holm)	6.1×10^{-8} (1.2×10^{-7})
Decision	reject H_0; smallest expected count 6,020

The side and the direction are dependent, but weakly: V is under a fifth of the smallest association that matters.

What the central limit theorem buys us

Test 1's interval relies on the CLT. Bounds from the lectures need weaker assumptions, at a price:

Interval	Assumes	95% interval for μ (bps)
t (CLT)	approx. normal day means	0.44 to 0.49
Chebyshev	finite variance only	0.40 to 0.53 (2.3× wider)
Hoeffding	$\lvert x\rvert \leq 51.8$ bps	-6.0 to 6.9 (228× wider)

Chebyshev, which assumes almost nothing, still excludes zero and still sits below the SDM. Hoeffding uses only the range, which the heavy tails make enormous, so it cannot even sign the effect.

Test 3 (P2): does a learned model see more than the imbalance?

a) Hypotheses, written before training any model

H_0	$\mu_\Delta \leq 0$
H_1	$\mu_\Delta > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	paired t-test over held-out days
Smallest difference that matters	ΔIC $= 0.01$

We train on days 0 to 240, stop early on days 241 to 300, and test once on the 180 days after that. A two-layer MLP sees 19 features of the auction state, including the recent path of prices and imbalance but not the stock's identity. Δ_d is the MLP's daily IC minus the imbalance's daily IC on the same test day. Each model is trained with five seeds and the predictions averaged.

b) Other tests we considered

Both ICs of a day are computed on the same stocks and moves, so they are paired.

Also considered	Why we did not use it as primary
Diebold-Mariano	compares squared forecast errors, not rankings; MLP vs a linear imbalance forecast gives $t = 16.9$, same decision
Sign-flip permutation	distribution-free check on Δ_d; p = 1.0×10^{-4}
LightGBM, GRU	alternative learners, not alternative tests: LightGBM IC 0.200, GRU 0.195; the MLP was fixed as primary in advance
z-test on all test rows	treats 359,960 rows as independent, ignoring the shared index move of a day

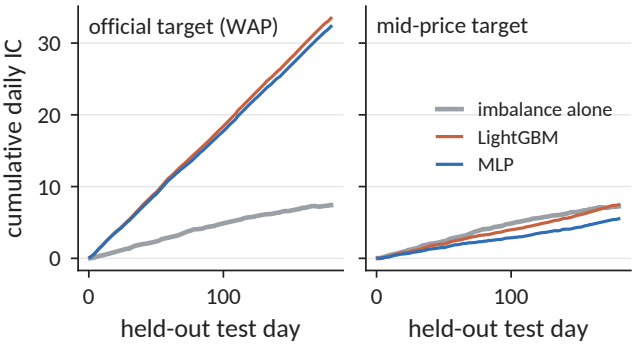


Figure 6: The learned gain is an artefact of how the target is built. Left: on the official target, which is computed from the size-weighted mid (WAP), both models leave the imbalance far behind. Right: on the plain mid-price move of the same rows, the imbalance alone is at least as good. Lines add up each test day's IC.

c) Result

Estimate	ΔIC $= +0.140$ (MLP 0.192, imbalance 0.052)
95% confidence interval	0.134 to 0.146
Test statistic (d.f.)	$t = 46.9$ (179)
p-value (Holm)	8.4×10^{-103} (3.4×10^{-102})
Decision	reject H_0

On the official target the MLP ranks next-minute moves 3.7 times better than the imbalance, and five seeds agree to within 0.0010 in IC. Figure 6 shows the catch.

The learned gain is WAP mechanics (check fixed before the five-seed run)

The official target is a return of the WAP, which shifts toward the thinner side of the book whenever depth changes. A model that reads depth can therefore predict the WAP without predicting the price; depth imbalance is indeed the model's most used feature (19% of LightGBM's split gain). Re-scored against the plain mid-price move, the MLP's IC falls to 0.031 while the imbalance keeps 0.040: the gain becomes -0.010 (95% CI -0.016 to -0.003). By the rule we fixed before the run, most of the learned gain is WAP mechanics. The imbalance itself passes the same check: its Test 1 edge is 0.31 bps on the WAP and 0.31 bps on the mid over the same rows.

Test 4 (P3): does 15:55 change how predictable the move is?

a) Hypotheses, written before running the test

H_0	$\mu_\Delta = 0$
H_1	$\mu_\Delta \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	paired t-test over days
Smallest difference that matters	$\lvert\Delta\text{IC}\rvert = 0.02$

The information coefficient (IC) is the rank correlation, across stocks, between signed imbalance and the next move: 0 means no predictive power, 1 a perfect ranking. Each day gives an IC before and after 15:55, and Δ_d is post minus pre. Two-sided, because the near price could absorb the imbalance information or sharpen it. A change of 0.02 is of the same order as the IC itself.

b) Other tests we considered

The two ICs of a day come from the same stocks in the same market, so they are paired; the paired t-test removes the day effect.

Also considered	Why we did not use it
Fisher z on two pooled correlations	assumes two independent samples, but they share stocks and days; gives $z = 20.5$
t-test for each correlation	tests each phase against zero, not the change; pre $r = 0.030$ ($t = 20.8$), post $r = 0.072$ ($t = 49.8$)
Sign test over days	distribution-free; 363 of 481 days rise, p = 4.8×10^{-30}
Sign-flip permutation	randomly swaps each day's pre and post, 10,000 times; largest 0.014, p = 1.0×10^{-4}

c) Result

Estimate	$\Delta\text{IC} = +0.047$ (IC 0.032 before, 0.080 after)
95% confidence interval	0.040 to 0.054
Test statistic (d.f.)	$t = 13.6$ (480)
p-value (Holm)	7.8×10^{-36} (2.3×10^{-35})
Decision	reject H_0

The imbalance predicts the next minute 2.5 times better after 15:55, and the whole interval lies beyond the threshold. We had predicted, in writing, a fall. The placebo test after Test 5 shows where the rise really comes from.

Test 5 (P3): is the next minute less volatile after 15:55?

a) Hypotheses, written before running the test

H_0	$\sigma^2_{\text{post}}/\sigma^2_{\text{pre}} = 1$
H_1	$\sigma^2_{\text{post}}/\sigma^2_{\text{pre}} \neq 1$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	F-test for two variances
Smallest difference that matters	ratio outside 0.90 to 1.11

A 10% change in variance is a 5% change in volatility, the scale at which a market maker would re-price quotes.

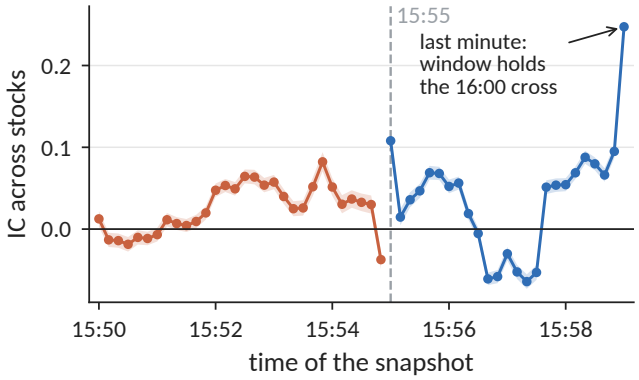


Figure 7: Predictability is not a step at 15:55: it spikes for one snapshot at 15:55, fades, turns negative near 15:57, and peaks in the last minute, whose window contains the 16:00 closing cross. Each point is the IC across stocks at one 10 s snapshot, averaged over days, with a 95% band.

b) Other tests we considered

The F-test is the syllabus test for two variances, but it assumes normal data and our target has excess kurtosis of 21 before and 24 after 15:55. We read it beside two checks that do not need normality.

Also considered	Why we did not use it as primary
Brown-Forsythe	robust to heavy tails; $W = 3{,}954$, p $< 10^{-300}$, same decision
t on daily log variance ratio	respects day dependence; ratio 0.79 (95% CI 0.77 to 0.80), $t = -24.5$
χ^2 for one variance	needs a known σ_0; we compare two unknown variances

c) Result

Estimate	$F = 0.771$ (sd 10.5 to 9.2 bps)
95% confidence interval	0.766 to 0.775
Test statistic (d.f.)	$F = 0.771$ (476,159, 476,159)
p-value (Holm)	$< 10^{-300}$ ($< 10^{-300}$)
Decision	reject H_0

The next minute is about 23% less variable after 15:55, and all three tests agree, including the two that do not assume normality. The change matters: the interval lies below 0.90.

Is 15:55 really the cause? Placebo boundaries (P3)

If the rule change drives Test 4, the IC should jump at 15:55 and nowhere else. The jump at 15:55 is real, $+0.057$ (95% CI 0.040 to 0.074), but the jump into the last minute is larger, $+0.193$ (0.181 to 0.206), so our pre-set criterion fails.

Figure 7 shows why. The IC leaps from -0.037 to 0.108 across 15:55, fades to -0.064 near 15:57, and reaches 0.247 for the last minute. Dropping that minute shrinks the post-minus-pre gain from 0.054 (0.047 to 0.060) to 0.014 (0.006 to 0.021), inside our threshold. Publishing the near price causes a brief spike; the lasting gain is the imbalance telling us about the closing print itself. We label this decomposition exploratory, since it came after the placebo result. The rise itself is positive in all 11 of 11 subgroups we fixed in advance (by period, size, spread and day volatility), and a classifier tells pre- from post-15:55 auction states apart with 82.3% accuracy on held-out days.

Test 6 (P4): can an outsider trade the imbalance?

a) Hypotheses, written before running the test

H_0	$p \leq 0.5$
H_1	$p > 0.5$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	one-proportion z-test
Smallest difference that matters	$p = 52\%$

p is the chance that a trade in the imbalance direction earns more than its own half-spread. A rule that wins 52% of the time is a two-point edge over a coin.

b) Other tests we considered

Also considered	Why we did not use it
Exact binomial	identical at $n = 738{,}083$; same decision
t on daily hit rates	dependence-robust check; $t = -133.0$
Mean net P&L	Test 1 already tests the mean; a rate is what a trader monitors

c) Result

Estimate	$\hat{p} = 39.2\%$ (52.0% before costs)
95% confidence interval	39.1 to 39.3%
Test statistic	$z = -185.4$
p-value (Holm)	1.000 (1.000)
Decision	fail to reject H_0

Before costs the rule wins slightly more than half the time; after paying the half-spread it wins only 39.2%, and the average trade loses -2.40 bps. On the mid price the hit rate is 37.3%, so the verdict does not depend on how the target is built.

Who can trade it? Edge against cost (P4)

Figure 8 sets the average edge per trade against the half-spread at every snapshot. The edge stays below the cost for the whole auction except the last minute. Trading only the strongest tenth of imbalances after 15:55 raises the gross edge to 1.77 bps, but those stocks quote a half-spread of 3.81, so the net is -1.99 bps (-2.18 to -1.79). Strong imbalances appear exactly where trading is expensive.

The learned model does not change this. Trading only when the MLP's forecast exceeds the half-spread appears to earn 0.68 bps per trade on the WAP target, but -0.40 bps (-0.51 to -0.28) on the mid price, a loss. At the 15:59 snapshot, and only there, the imbalance edge of 2.40 bps beats a half-spread of 1.92, a net 0.48 bps (0.39 to 0.58) that shrinks from 0.74 in the first half of the days to 0.23 in the second. We found it after looking, and it ignores the cost of hedging the index and of getting filled, so it is a lead for future work, not a result.

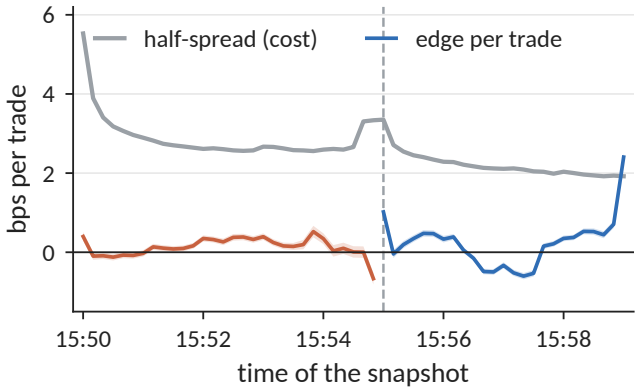


Figure 8: The imbalance edge (coloured, with 95% band) never covers the half-spread (grey) until the last minute. Each point averages, over days, the gross edge of trading in the imbalance direction at that snapshot.

Answers

P1	Yes. The imbalance predicts the next minute on 458 of 481 days, in real price moves, not only in the WAP.
P2	Only on paper. The learned gain on the official target is WAP mechanics; on the mid price the imbalance alone is as good.
P3	Partly. Predictability and volatility change after 15:55, but the lasting gain is the closing cross, not the rule.
P4	No, except possibly in the last minute, an exploratory lead. The information is in the price to within trading costs.

6. Limitations

- **Two rules change at once, next to the close.** At 15:55 the near price appears and market-on-close entry closes; the placebo test shows the post-15:55 gain is mostly the final minute, so time to the close and the rule change are not separated.
- **Association, not cause.** The imbalance may predict because the same traders move both, not because traders read it.
- **Days treated as independent.** Market-wide news can link consecutive days; the effects are large enough that modest dependence would not change a decision.
- **One model family, one split.** P2 uses one MLP design and one chronological split; a different architecture could find real price information we did not.
- **Costs are approximate.** The half-spread is a snapshot quote; fees, impact, index hedging and auction fills are not modelled, so the true cost is higher.
- **One exchange, anonymised period.** Dates and tickers are hidden, so results cover these 200 Nasdaq stocks over about two years.

Supplementary material (figures and tables for every check above, the full pre-registration log and the model settings) is in `report/supplementary.pdf` in the code repository.